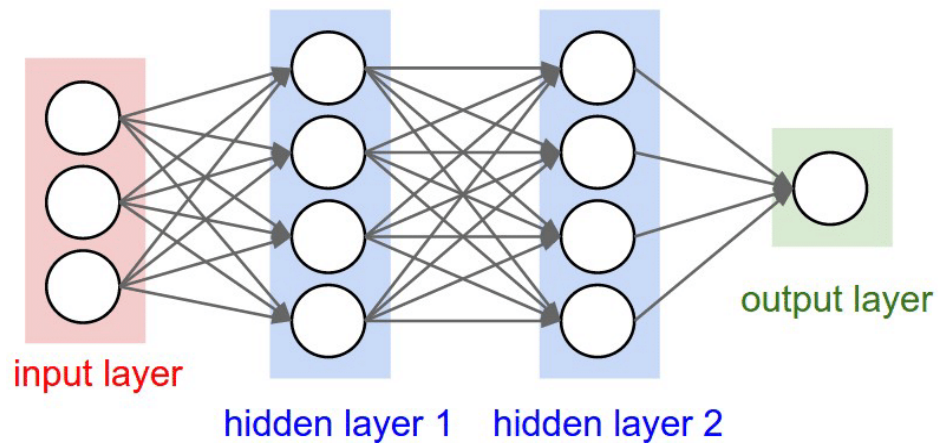


Multi-layer Perceptron (MLP)

- These networks have one or more hidden layers between the input and output layers, allowing them to capture more complex patterns.



Important technical terms

- **Batch Size:**

The number of samples (rows) processed **at once** before updating the model.

- **Epoch:**

One **epoch** means the model has seen **all** the training data **once**.

- **Iteration (Steps per Epoch):**

One **iteration** means the model has processed **one batch** of data.

The number of **iterations per epoch** depends on the total dataset size.

- **Early Stopping:**

Training can be stopped early if the model's performance on the validation set stops improving.

Commonly Used Activation Functions:

1. Sigmoid:

- Output: (0,1)
- Suitable for binary classification but can suffer from vanishing gradients.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

2. Tanh:

- Output: (-1,1)
- Centered around zero, which can help in optimization compared to sigmoid.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

3. ReLU (Rectified Linear Unit):

$$\text{ReLU}(x) = \max(0, x)$$

- Output: [0,∞)
- Efficient and widely used in hidden layers.
- Can suffer from "dying neurons" where gradients become zero.

4. Leaky ReLU:

- Output: [0.01x,∞)
- A variant of ReLU that allows a small gradient for negative inputs, mitigating the dying neuron problem.

5. Softmax:

- Used in the output layer for multi-class classification.
- Converts logits into probabilities.

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$